

Seongyoon (Sean) Kim

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EDUCATION

Ph.D. in Mathematics	University of Illinois Urbana-Champaign Expected Dec 2027
M.S. in Mathematics	University of Illinois Urbana-Champaign May 2022
B.A. in Economics	Sungkyunkwan University Aug 2019
B.S. in Economics and Business Economics	Erasmus University Rotterdam Aug 2018

QUANTITATIVE SKILLS

- **Math & Stat:** Stochastic Modeling | Time-Series Modeling | Predictive Modeling | Optimization | Regression Analysis
- **AI/ML:** Neural ODEs | Sequence Models | Gradient-Boosted Trees | Rare-Event Classification | Time-Series ML
- **Programming & Tools:** Python | C++ | R | Stata | MATLAB | Mathematica | SQL

SELECTED RESEARCH & PUBLICATIONS

Predicting Price Surges with Neural ODE	May 2026
<i>Manuscript under review at a major machine learning conference [Link]</i>	
• Designed a classification-oriented Neural ODE framework for rare price-surge prediction and a generalized jump-diffusion benchmark capturing dependence among market jumps. • Reduced overfitting through continuous-time latent refinement and improved surge-prediction performance by training directly for classification rather than future-trajectory reconstruction. • Outperformed neural and stochastic benchmarks with Neural ODE on both KOSDAQ and IWM, improving AP over LSTM by approximately 35% and 50.6%, and increasing recall from 62.0% to 70.1% and from 57.0% to 73.4%, respectively.	
Price-Surge Prediction for Systematic Trading	Mar 2026–Present
<i>Work in progress [Link]</i>	
• Constructed a portfolio framework that transforms price-surge forecasts into systematic trading decisions, targeting improved trade selection and portfolio performance rather than prediction accuracy in isolation. • Addressed the gap between predictive performance and realizable trading outcomes through first-passage-time labels and execution-aware backtests with transaction costs, market impact, position limits, and barrier exits. • Achieved over 60% 20-day surge-prediction accuracy with XGBoost and approximately 80% trade win rates in portfolios constructed from stocks with model prediction scores of 0.8–0.9 under 5–8% take-profit targets on U.S. test data.	
On the Expansion of Risk Pooling	Jan 2026
<i>Management Science, Articles in Advance (2026) [Link]</i>	
• Formulated a risk-pooling framework using entropic risk measures and reinsurance to characterize participation incentives and consensus under pool expansion across alternative pricing rules. • Provided a decision framework for determining when pool expansion is mutually beneficial, showing how pricing and reinsurance choices affect member participation and pool stability. • Applied the framework to support pool-expansion and reinsurance decisions through two real-data case studies drawing on the Caribbean Catastrophe Risk Insurance Facility and insurance-product records.	
Tokenization of Distributed Insurance by Auction	Aug 2024
<i>Japanese Journal of Statistics and Data Science, Vol. 7, pp. 1039–1057 (2024) [Link]</i>	
• Developed an auction-based framework for tokenizing insurance portfolios, using a knapsack algorithm to optimize investment allocation and return distribution in decentralized finance markets. • Demonstrated with real-world data that the proposed mechanism improves trading efficiency relative to a naive capital-payment minimization approach and analyzed its performance under strategic bid merging and splitting.	

LEADERSHIP & SERVICE

Research Lead , Financial Research in AI, Mathematics, and Economics (FRAME)	2024–Present
• Built and led an interdisciplinary quantitative research team, recruiting mathematics and mechanical engineering Ph.D. researchers and data scientists and assigning roles across modeling, implementation, validation, and analysis. • Derived core mathematical and machine-learning models and oversaw the end-to-end research process, including data analysis, backtesting, portfolio evaluation, and integration of individual contributions into final research outputs.	
Sergeant , Republic of Korea Army	2025–2026
Journal Peer Reviewer , <i>Risk Sciences</i>	2024
Graduate Research & Teaching Assistant , University of Illinois Urbana-Champaign	2020–2024; 2026–Present

AWARDS & PRESENTATIONS

Research Fellow , Predictive Analytics & Risk Management, University of Illinois Urbana-Champaign	Jun 2024
• Awarded a \$5,500 research fellowship.	
Selected Presentations: Int’l Conference on Insurance & Actuarial Science; INFORMS Annual Meeting	Jun 2024; Oct 2022
Grand Prize, 3rd Financial New Product Idea Competition , Hyundai Motor Securities	Jul 2019
• Awarded a KRW 5,000,000 cash prize.	